DATA WAREHOUSE IMPLEMENTATION PROJECT

Project Objective

To design and implement an **end-to-end data pipeline** that ingests data from an **OLTP source system (SQL Server)** into a **Snowflake-based Data Warehouse**, ensuring full support for historical data tracking (SCD Type 2) and preparing the data for **business analytics, reporting, and visualization**.

Project Workflow

1. Source System Setup (OLTP)

- Restored data from a *.bak* file into **SQL Server (SSMS)** to replicate a transactional database environment.
- Analyzed the source structure to identify required tables for downstream processing.

2. ETL Pipeline Development (SQL Server → Snowflake)

- Built a data pipeline in **Python** to extract data from SQL Server and load it into **Snowflake**.
- Designed a two-layer schema architecture in Snowflake:
  - **Raw Schema (Sales_Temp):** Captures data directly from SQL Server.
  - **Staging Schema (Sales):** Serves as an intermediate layer for data cleansing and preparation.

3. Change Data Capture (CDC) & Historical Tracking

- Implemented **Slowly Changing Dimension Type 2 (SCD2)** to manage historical data.
- Added control columns to the staging tables, including:
  - IsUpdate, IsDelete, IsInsert, IsCurrent
  - StartDate, EndDate
- Developed **Snowflake Stored Procedures** to automatically detect and process **Insert, Update, and Delete** operations.

4. Data Warehouse Design (OLAP Layer)

- Designed a **Star Schema model** consisting of:
    - **Fact Tables:** Storing business metrics and transactional measures.
    - **Dimension Tables:** Storing attributes, hierarchies, and historical context.
- Transformed data from the staging layer (**OLTP structure**) into the Data Warehouse layer (**OLAP structure**) optimized for analytics.

5. Reporting & Visualization Preparation

- Delivered a clean and integrated dataset for **BI tools (Power BI / Tableau)**.
- Ensured support for **multi-dimensional reporting, trend analysis, and business performance tracking**.

Key Outcomes

- Delivered a fully automated **end-to-end data pipeline**: SQL Server → Snowflake (Raw & Staging) → Data Warehouse.
- Enabled reliable **historical data tracking** using CDC and SCD Type 2, ensuring auditability and version control.
- Successfully designed a **scalable Data Warehouse layer** supporting analytical queries and business reporting.
- Established a strong foundation for **BI dashboards and advanced analytics use cases**.